

# ASHLEY (YE) GAO

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📍 540 Landrum Dr, Williamsburg, VA 23185

## EMPLOYMENT

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**William & Mary**

2023 - Present

*Assistant Professor in Computer Science*

## EDUCATION

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**University of Virginia**

2019 - 2023

Advisor: John A. Stankovic

Thesis: *Addressing Realisms Faced by Deep Learning Models in Cyber Physical Systems*

*Ph.D. in Computer Science*

**University of Virginia**

2017 - 2019

*M.S. in Computer Science*

**University of California, San Diego**

2012 - 2017

*B.S. in Computer Science*

*B.A. in Literatures of the World*

## RESEARCH STATEMENT

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My research lies at the intersection of **affective computing**, **audio-language modeling**, and **robust machine learning**. I develop methods for **emotion-aware representation learning and generation**, with a focus on audio large language models, controllable text-to-speech, and speech emotion recognition. My work combines advances in **representation learning**, **generative modeling** with applications in **computational healthcare** and **human-centered AI**.

## SERVICES

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**Editorial Roles**

**S5. Associate Editor.** Elsevier Smart Health.

2024 - Present

**Grant Review Panels (Invited)**

**S4. Panelist.** National Institutes of Health (NIH).

2025

**S3. Panelist.** National Institutes of Health (NIH).

2024

**S2. Panelist.** National Science Foundation (NSF).

2024

**Conference Leadership**

**S1. Demo & Poster Chair.** IEEE/ACM CHASE (Connected Health: Applications, Systems and Engineering Technologies).

2024

## SELECTED ACCEPTED PUBLICATIONS

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**Conference and Journal Publications**

[ICML'26] Du, H., Shi, J., Lu, S., Zhou, G., **Gao, Y.**, "Sparse Autoencoders for Interpretable Emotion Control in Text-to-Speech." Proceedings of the International Conference on Machine Learning (ICML), 2026.

[ACL'26] Nair, T., Nafee, M. W., Jiang, M., **Gao, Y.**, Chen, H., Zhang, Y., "Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing." Findings of the Association for Computational Linguistics (ACL), 2026.

- [**ACL'26**] Shi, J., Du, H., Song, X., Hong, Y. A., Zhang, Y., Gao, Y., "AffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling." Findings of the Association for Computational Linguistics (ACL), 2026.
- [**ICASSP'26**] Shi, J., Du, H., He, Y., Hong, Y. A., Gao, Y., "Emotion-Aligned Generation in Diffusion Text-to-Speech Models via Preference-Guided Optimization." IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., Gao, Y., "EMO-TTA: Improving Test-Time Adaptation of Audio-Language Models for Speech Emotion Recognition." IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., Gao, Y., "Plug-and-Play Emotion Graphs for Compositional Prompting in Zero-Shot Speech Emotion Recognition." IEEE ICASSP, 2026.
- [**ICASSP'26**] Mersa, O., Jiang, M., Gao, Y., Li, Q., Zhang, Y., "Hierarchical Convolution Multibranch Transformer for EEG Signals." IEEE ICASSP, 2026.
- [**ICASSP'26**] Jiang, M., Chen, Y., Liu, G., Sahu, A., Gao, Y., Zhang, Y., "Pairs, Not Labels: Predicting Protein-Phenotype Associations via Link Prediction." IEEE ICASSP, 2026.
- [**EMNLP'25**] Du, H., Shi, J., Myerston, J., Lu, S., Zhou, G., Gao, Y., "Role-Guided Annotation and Prototype-Aligned Representation Learning for Historical Literature Sentiment Classification." Findings of EMNLP, 2025.
- [**InterSpeech'25**] Du, H., Lu, S., Zhou, G., Gao, Y., "Emotion-Aware Audio Large Language Models with Dual Cross-Attention and Context-Aware Instruction Tuning." InterSpeech, 2025.
- [**InterSpeech'25**] Shi, J., Zhang, Y., Gao, Y., "Contrastive Learning for Speech Emotion Domain Generalization via Soft Prompt Tuning." InterSpeech, 2025.
- [**CHASE'25**] Clapham, J., Zhou, M., MacDonald, C., Koltermann, K., Gao, Y., Shao, H., "ElectroMeter: The Practical Electrolyte Measurement System." IEEE/ACM CHASE, 2025.
- [**JMIR'25**] Ko, E., Gao, Y., Wang, P., Wijayasingha, L., Wright, K. D., Gordon, K. C., Wang, H., Stankovic, J. A., & Rose, K. M. "Challenges to recruiting dementia caregiving dyads in community-based settings." JMIR Formative Research, 2025.
- [**TMLR'24**] Macdonald, C., Chu, Z., Stankovic, J., Zhou, G., Shao, H., Gao, Y., "MiddleGAN: Generating Inter-domain Images with GANs." Transactions on Machine Learning Research, 2024.
- [**CHASE'24**] Koltermann, K., Clapham, J., Blackwell, G., Jung, W., Burnet, E., Gao, Y., Shao, H., et al., "Gait-Guard: Turn-aware Freezing of Gait Detection for Non-intrusive Intervention Systems." IEEE/ACM CHASE, 2024. (**Best Paper Award**)
- [**SmartComp'23**] Gao, Y., Baucom, B., Gordon, K., Rose, K., Wang, H., Stankovic, J. "E-ADDA: Unsupervised Adversarial Domain Adaptation Enhanced by a New Mahalanobis Distance Loss." IEEE SmartComp, 2023.
- [**HEALTH'21**] Gao, Y., Salekin, A., Gordon, K., Rose, K., Wang, H., Stankovic, J. "Emotion Recognition Robust to Indoor Environmental Distortions Using Out-of-Distribution Detection." ACM Transactions on Computing for Healthcare, 2021.
- [**PERCOM'21**] Gao, Y., et al. "Out-of-the-Box Deployment to Support Research on In-Home Care of Alzheimer's Patients." IEEE Pervasive Computing, 2021.
- [**JAN'21**] Rose, K. M., et al., Gao, Y., "Smarthealth technology study protocol for dementia caregiving relationships." Journal of Advanced Nursing, 2021.
- [**SENSYS'20**] Gao, Y., et al. "A Monitoring and Recommendation System for In-Home Caregivers (Demo)." ACM SenSys, 2020.

Full publication list: <https://scholar.google.com/citations?user=XvbRi5wAAAAJ&hl=en>

## THE GALATEA LAB

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### **Mentoring Ph.D. Students**

**M3.** Jiacheng Shi, Ph.D. Student, William & Mary.  
First author publications: ICASSP, ACL, Interspeech 2024 - Present

**M2.** Hongfei Du, Ph.D. Student, William & Mary.  
First author publications: ICML, ACL, EMNLP, Interspeech 2024 - Present

### **Mentoring Undergraduate Students**

**M1.** Lilly Liu, Undergraduate Student, William & Mary.  
Initial placement: Carnegie Mellon University (CMU). 2024 - Present

### ***Selected Research Directions***

**D3.** Domain-Generalized Audio Sensing — robust speech emotion recognition under distribution shift

**D2.** Sentiment Steering in LLMs — controllable generation of emotional tone in language models

**D1.** Audio Stress Detection for ADRD Caregivers — mobile and scalable systems for caregiver support ([Linda Gaw][1])

## HONORS AND AWARDS

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### ***Research Awards***

**H7.** Best Paper Award, IEEE/ACM CHASE. 2024

**H6.** Best Paper Runner-Up Award, IEEE SmartComp. 2023

**H5.** Best Poster Award, Computer Science Symposium, University of Virginia. 2022

### ***Service Awards***

**H4.** Best Service Award, IEEE/ACM CHASE. 2024

### ***Fellowships and Academic Honors***

**H3.** Ph.D. Fellowship, Computer Science, University of Virginia. 2019

**H2.** Academic Excellence Fellowship, Computer Science, University of Virginia. 2017

**H1.** Honors with High Distinction, University of California, San Diego. 2016

## REVIEWER SERVICE

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**R6. Primary Reviewer.** NeurIPS. 2026

**R5. Primary Reviewer.** ICML, AAAI, ICLR. 2025

**R4. Primary Reviewer.** NeurIPS, AAAI, ICLR, ACM Health. 2024

**R3. Primary Reviewer.** AAAI, NeurIPS, ICLR. 2023

**R2. Primary Reviewer.** IEEE Transactions on Affective Computing; AAAI; Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp); NeurIPS. 2022

**R1. Primary Reviewer.** ACM Transactions on Computing for Healthcare. 2021

## PROFESSIONAL MEMBERSHIPS

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**P2. Association for Computing Machinery (ACM).** Member. 2025 - Present

**P1. Institute of Electrical and Electronics Engineers (IEEE).** Member. 2025

## PUBLISHED CREATIVE WRITING

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### ***Books***

**C4.** *The Tempest's Call*. Pegasus Publishers, United Kingdom. (To Appear) 2026

**C3.** *The Celestial Blade*. Pink Hydra Press, South Africa. (To Appear) 2026

### **Short Fiction**

- C2.** *The Purpose of a Name*. Neon Dystopia, United States. (To Appear) 2026  
**C1.** *The Mother of Gaia*. Bewildering Stories, Canada. 2025

### FEATURED

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- F1.** Honors student of the Department of Literature at UCSD. [Webpage] 2016

### REFERENCES

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#### **John A. Stankovic**

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